

Technical Appendix: Sampling-Boundary Distortion in Sharpe-Ratio Evidence from Financial Panels

1 Purpose

This appendix provides derivations, calibration checks, robustness tables, and reproducibility artifacts omitted from the main manuscript for brevity.

2 Mean-Return Design-Effect Proof

For a selected row model $y_{t,i} = \mu + \lambda_t + \epsilon_{t,i}$ with constant selected count $\bar{n}$, same-date intraclass correlation ρ , and independent dates, the row-pooled mean equals the average of date means:

$$\hat{\mu} = \frac{1}{T\bar{n}} \sum_{t=1}^T \sum_{i=1}^{\bar{n}} y_{t,i} = \frac{1}{T} \sum_{t=1}^T \bar{y}_t.$$

The row-naive variance report is $\sigma^2/(T\bar{n})$. The true one-date variance is

$$\text{Var}(\bar{y}_t) = \frac{1}{\bar{n}^2} \{ \bar{n}\sigma^2 + \bar{n}(\bar{n} - 1)\rho\sigma^2 \} = \frac{\sigma^2}{\bar{n}} \{ 1 + (\bar{n} - 1)\rho \}.$$

Thus

$$\frac{se_{date,true}}{se_{row,naive}} = \sqrt{1 + (\bar{n} - 1)\rho}.$$

This is the Moulton-like design effect used as the mean-return special case in the main text. It is a closed-form special case, not the full Sharpe-ratio distortion result.

3 Sharpe UVIF Derivation

These calculations supply the algebra behind Proposition 1 in the main manuscript. Let R_t be the date-level portfolio excess return and $G_t = (R_t, R_t^2)'$. The annualized Sharpe ratio is the smooth functional

$$f(\mu, m_2) = \sqrt{A_f} \frac{\mu}{(m_2 - \mu^2)^{1/2}}, \quad d_R = \nabla f(\mu_R, m_{2,R}).$$

The annualization factor A_f is source-specific: the public registry uses 252 for daily sources and 12 for monthly sources. On sets where $v_R = m_{2,R} - \mu_R^2 > 0$, the first-order expansion is

$$\widehat{\text{SR}}_R - \text{SR}_R = d'_R(\hat{\theta}_R - \theta_R) + o_p(T_R^{-1/2}), \quad \theta_R = \mathbb{E}[G_t].$$

With long-run covariance $\Omega_R = \Gamma_0^R + \sum_{k \geq 1} (\Gamma_k^R + \Gamma_k^{R'})$,

$$\text{Var}_{LR}(\widehat{\text{SR}}_R) = T_R^{-1} d'_R \Omega_R d_R + o(T_R^{-1}).$$

The row-pooled IID calculation applies the same delta method to $\bar{n}T_R$ row observations $H_{t,i} = (Y_{t,i}, Y_{t,i}^2)'$:

$$\text{Var}_{IID, \text{row}}(\widehat{\text{SR}}_{\text{row}}) = (\bar{n}T_R)^{-1} d'_Y \Psi_Y d_Y + o((\bar{n}T_R)^{-1}).$$

Taking the ratio gives the Sharpe UVIF:

$$\text{UVIF}_{\text{SR}} = \frac{\bar{n} d'_R \Omega_R d_R}{d'_Y \Psi_Y d_Y} = \frac{\bar{n} d'_R \Gamma_0^R d_R}{d'_Y \Psi_Y d_Y} \times \frac{d'_R \Omega_R d_R}{d'_R \Gamma_0^R d_R}.$$

The first term is the cross-sectional component and the second term is the serial long-run factor. Componentwise,

$$d'_R \Omega_R d_R = \frac{A_f}{v_R^3} \left[m_{2,R}^2 \Omega_{11} - \frac{\mu_R m_{2,R}}{2} (\Omega_{12} + \Omega_{21}) + \frac{\mu_R^2}{4} \Omega_{22} \right].$$

The Ω_{22} term contains long-run covariances of squared returns and therefore brings fourth moments into Sharpe-ratio variance.

4 Optional Capacity-Stress Note

The public sorted-portfolio examples in the main manuscript do not contain average daily volume or order-size information, so they do not support a capacity claim. For traded-security panels where those inputs are available, a square-root impact diagnostic can be added as an optional extension to the statistical claim boundary.

Proposition 1 (Square-root impact sensitivity bound). *Suppose a traded panel has row-pooled gross mean $\bar{r}_{\text{gross}}$, row-pooled return standard deviation σ_{row} , strictly positive average turnover $\widehat{\text{TO}}$, T trading dates, square-root impact cost $\lambda \widehat{\text{TO}} \sqrt{A/V}$, and a dependence standard-error inflation factor $\mathcal{M} \geq 1$ relative to row-pooled inference. For a one-sided*

level- α sensitivity margin, any statistically screened asset scale A^* must satisfy

$$A^* \leq V \left(\frac{\bar{r}_{\text{gross}} - z_\alpha \mathcal{M} \sigma_{\text{row}} / \sqrt{T}}{\lambda \widehat{\text{TO}}} \right)^2 \quad (1)$$

when the numerator is positive; otherwise no positive scale is screened by the margin.

For a square-root impact diagnostic, the dependence-adjusted one-sided lower confidence margin for gross mean return is

$$\bar{r}_{\text{gross}} - z_\alpha \mathcal{M} \sigma_{\text{row}} / \sqrt{T},$$

where σ_{row} is the standard deviation of row-pooled returns before date aggregation or UVIF adjustment. Square-root impact subtracts $\lambda \widehat{\text{TO}} \sqrt{A/V}$. Requiring the net lower margin to be nonnegative gives

$$\lambda \widehat{\text{TO}} \sqrt{A/V} \leq \bar{r}_{\text{gross}} - z_\alpha \mathcal{M} \sigma_{\text{row}} / \sqrt{T}.$$

Rearranging yields the sensitivity bound in Proposition 1. If the volatility input is already the date-level portfolio volatility, the date boundary has already embedded the cross-sectional UVIF component, so the cross-sectional part of $\mathcal{M}$ must not be multiplied in again.

5 Higher-Moment UVIF and ERIR Details

For $\theta^{(4)} = (\mu, m_2, m_3, m_4)'$, define skewness and kurtosis from the central moments implied by $\theta^{(4)}$. Any adjusted Sharpe functional $h(\theta^{(4)})$ has a delta-method variance $\nabla h' \Omega^{(4)} \nabla h$, where $\Omega^{(4)}$ is the long-run covariance matrix of $(R_t, R_t^2, R_t^3, R_t^4)'$. The higher-moment UVIF is the corresponding ratio of date-level long-run variance to row-pooled IID variance. It is a diagnostic extension, not an additional empirical rejection gate.

The Effective Row Information Retention (ERIR) diagnostic is a design-effect retention calculation, not a hypothesis test or a formal information claim. Under the same equicorrelated approximation used for the mean-return design effect,

$$DE_{\mu, xs} = 1 + (\bar{n} - 1)\rho, \quad \text{ERIR} = \{1 + (\bar{n} - 1)\rho\}^{-1}.$$

Thus an ERIR of 0.111 means that roughly $1 - 0.111 = 0.889$, or 89 percent, of nominal row-level information is redundant under this diagnostic approximation. Confirmatory inference remains based on date-level HAC-delta and dependent-bootstrap procedures. When the descriptive equicorrelation estimate is non-positive, the unclipped ERIR can exceed one and the corresponding UVIF can fall below one; the main manuscript dis-

plays a conservative one-sided UVIF floor at one and reports the unclipped diagnostics separately.

6 Synthetic Positive Control

This section gives the appendix version of the known-edge calibration check. The synthetic panel is not empirical evidence for a public strategy; it verifies that the composite audit gate can accept a declared positive signal when the data-generating process contains one. The generated panel has $T = 5000$ dates and $N = 100$ names. Signed returns include daily alpha $0.12/252$, idiosyncratic volatility 0.01 , a common loading of 0.50 , AR(1) common-factor coefficient 0.10 , signal-stay probability 0.98 , and seed 777 . The displayed run is one representative known-edge calibration with 5000 bootstrap draws and 10000 same-date permutations; costs use the same 5 bps turnover-stress convention as the public artifacts. The replication screen below uses fewer resampling draws and is reported only as a power sanity check. The data-generating process is

$$\begin{aligned} f_t &= 0.10f_{t-1} + u_t, & \epsilon_{t,i} &\sim N(0, 0.01^2), \\ z_{t,i} &\in \{-1, 1\}, & \Pr(z_{t,i} = z_{t-1,i}) &= 0.98, \\ r_{t+1,i} &= z_{t,i}(0.12/252) + 0.50f_t + \epsilon_{t,i}, & s_{t,i} &= z_{t,i}. \end{aligned}$$

The audit then applies the same signed-row construction, date aggregation, HAC-delta inference, Romano-Wolf menu adjustment, same-date permutation, and 5 bps cost stress used in the public artifacts.

Table 1: Synthetic positive-control evaluation.

Panel	$\bar{\rho}$	SE infl.	SR	HAC p^+	RW p	Perm. p^+
Synthetic control	0.202	4.58	1.744	7.67e-13	2.00e-04	1.00e-04

Table 2: Positive-control replication screen.

Reps	Boot	Perms	Block pass	HAC pass	RW pass	Perm. pass	Cost pass	All pass
100	499	199	100.0%	100.0%	100.0%	100.0%	100.0%	100.0%

Note: The replication screen uses fewer resampling draws than the detailed calibration row and is reported only as a power sanity check.

7 Full Empirical Artifacts

These tables are audit logs rather than primary evidence; the main text reports the compressed interpretation. The inferential tables in this appendix report only public sources

that loaded and produced reproducible artifacts. Failed source-access attempts remain in the generated provenance files and campaign metadata, but are not treated as empirical evidence. Panel candidates and pre-aggregated factor benchmarks are reported in separate groups because only the former provide row-level signals for same-date permutation and sampling-boundary diagnostics. For each empirical table group, the check is whether the result survives moving from rows to date returns, whether nearby HAC and dependent-bootstrap variants agree, and whether placebo or cost diagnostics overturn the simple Sharpe interpretation. Large disagreement across these tables is a robustness warning, not a new discovery channel.

This supplement reports loaded sources in separated groups. The full-campaign loader provenance is retained in the generated metadata; it is not a comparable pass/fail audit table. The tables are audit logs rather than primary evidence; the main text reports the compressed interpretation.

Table 3: Source frequency and annualization metadata.

Source	type	period	A_f	dates
AQR Betting Against Beta, equity factors	single_series_factor	daily	252.000	25281
AQR HML Devil	single_series_factor	daily	252.000	26590
AQR Quality Minus Junk	single_series_factor	daily	252.000	17681
AQR Value and Momentum Everywhere	single_series_factor	monthly	12.000	651
French momentum deciles, rank-threshold panel	panel	daily	252.000	26131
French WML decile benchmark	panel	daily	252.000	26131
French 25 Size/BM, skipped-momentum panel	panel	daily	252.000	3477

Table 4: Panel candidates subject to the composite audit gate.

Panel candidate	SR	HAC p+	Boot p+	RW p	Perm p	Net SR	Audit gate 5%	Applicable checks 5%
French momentum deciles, rank-threshold panel	0.447	7.54e-05	2.00e-04	2.00e-04	N/A	0.447	no	yes
French WML decile benchmark	0.497	5.53e-06	2.00e-04	2.00e-04	1.00e-04	0.497	yes	yes
French 25 Size/BM, skipped-momentum panel	0.058	0.412	0.422	0.480	0.943	-0.050	no	no

Table 5: Single-series factor benchmarks subject to time-series inference only.

Single-series benchmark	SR	HAC p+	Boot p+	RW p	Scope
AQR Betting Against Beta, equity factors	0.741	3.08e-13	2.00e-04	2.00e-04	time-series only
AQR Quality Minus Junk	0.536	2.26e-05	2.00e-04	2.00e-04	time-series only
AQR HML Devil	0.336	0.004	0.002	0.002	time-series only
AQR Value and Momentum Everywhere	0.572	1.62e-04	2.00e-04	2.00e-04	time-series only

Note: Time-series only means not eligible for same-date permutation, row-retention diagnostics, or the composite panel audit gate.

Table 6: Canonical French momentum benchmark validation.

Candidate	thr	n	start	end	panel SR	direct WML SR	scale
French WML decile benchmark	0.900	26131	1926-11-04	2026-04-30	0.497	0.497	0.500

Table 7: Selected-count and dependence diagnostics for panel candidates.

Candidate	status	thr	rows	dates	avg names	rho	SE infl.
French momentum deciles, rank-threshold panel	Available	0.700	104524	26131	4.000	-0.138	0.766
French WML decile benchmark	Available	0.900	52262	26131	2.000	-0.583	0.646
French 25 Size/BM, skipped-momentum panel	Available	0.500	45201	3477	13.000	0.667	3.000

Table 8: Row-naive and date-level p-value diagnostics for panel candidates.

Candidate	thr	row t	HAC z	row p+	HAC p+	RW p
French momentum deciles, rank-threshold panel	0.700	3.584	3.790	1.70e-04	7.54e-05	2.00e-04
French WML decile benchmark	0.900	3.299	4.395	4.85e-04	5.53e-06	2.00e-04
French 25 Size/BM, skipped-momentum panel	0.500	0.664	0.222	0.253	0.412	0.480

Table 9: Unclipped and one-sided floored UVIF diagnostics for panel candidates.

Candidate	rho	SE infl.	UVIF unclipped	UVIF displayed	ERIR unclipped	ERIR displayed
French momentum deciles, rank-threshold panel	-0.138	0.766	0.587	1.000	1.704	1.000
French WML decile benchmark	-0.583	0.646	0.417	1.000	2.399	1.000
French 25 Size/BM, skipped-momentum panel	0.667	3.000	9.002	9.002	0.111	0.111

Note: Displayed main-text UVIF values are floored at one when the equicorrelation diagnostic is non-inflationary; unclipped values remain descriptive diagnostics and are not rejection rules.

Table 10: Dependence audit summary for panel candidates.

Candidate	Row p+	UVIF	Audit p+	Net SR	Status
French momentum deciles, rank-threshold panel	1.70e-04	1.000	2.00e-04	0.447	Permutation uninformative; applicable checks pass
French WML decile benchmark	4.85e-04	1.000	2.00e-04	0.497	Passes composite gate
French 25 Size/BM, skipped-momentum panel	0.253	9.002	0.943	-0.050	Fails statistical and cost gates

Note: Displayed UVIF is a one-sided inflation summary floored at one. Degenerate same-date permutation diagnostics are reported as uninformative rather than converted into failures.

Table 11: Horizon-effect audit for the dynamic Size/BM momentum panel.

status

Not available

Table 12: Primary-threshold dependence-aware Sharpe inference.

Candidate	thr	Method	SR	SE	CI low	CI high	p+	n
AQR Betting Against Beta, equity factors	0.000	block	0.741	0.106	0.533	0.951	2.00e-04	25281
AQR Betting Against Beta, equity factors	0.000	HAC	0.741	0.103	0.540	0.943	3.08e-13	25281
AQR Betting Against Beta, equity factors	0.000	prewhite	0.741	0.104	0.538	0.945	4.38e-13	25281
AQR HML Devil	0.000	block	0.336	0.114	0.105	0.552	0.002	26590
AQR HML Devil	0.000	HAC	0.336	0.126	0.089	0.583	0.004	26590
AQR HML Devil	0.000	prewhite	0.336	0.111	0.119	0.553	0.001	26590
AQR Quality Minus Junk	0.000	block	0.536	0.133	0.276	0.795	2.00e-04	17681
AQR Quality Minus Junk	0.000	HAC	0.536	0.131	0.279	0.794	2.26e-05	17681
AQR Quality Minus Junk	0.000	prewhite	0.536	0.128	0.285	0.787	1.40e-05	17681
AQR Value and Momentum Everywhere	0.000	block	0.572	0.156	0.268	0.882	2.00e-04	651
AQR Value and Momentum Everywhere	0.000	HAC	0.573	0.159	0.261	0.885	1.62e-04	651
AQR Value and Momentum Everywhere	0.000	prewhite	0.573	0.158	0.263	0.882	1.42e-04	651
French momentum deciles, rank-threshold panel	0.700	block	0.447	0.114	0.222	0.667	2.00e-04	26131
French momentum deciles, rank-threshold panel	0.700	HAC	0.447	0.118	0.216	0.679	7.54e-05	26131
French momentum deciles, rank-threshold panel	0.700	prewhite	0.447	0.114	0.224	0.670	4.18e-05	26131
French WML decile benchmark	0.900	block	0.497	0.111	0.275	0.708	2.00e-04	26131
French WML decile benchmark	0.900	HAC	0.497	0.113	0.275	0.718	5.53e-06	26131
French WML decile benchmark	0.900	prewhite	0.497	0.111	0.279	0.715	3.99e-06	26131
French 25 Size/BM, skipped-momentum panel	0.500	block	0.058	0.261	-0.483	0.540	0.422	3477
French 25 Size/BM, skipped-momentum panel	0.500	HAC	0.058	0.260	-0.452	0.567	0.412	3477
French 25 Size/BM, skipped-momentum panel	0.500	prewhite	0.058	0.255	-0.442	0.557	0.410	3477

Table 13: Same-date signal-permutation diagnostics for panel candidates.

Candidate	thr	obs SR	null mean	null sd	p+	status	perms
French momentum deciles, rank-threshold panel	0.000	0.395	0.410	0.039	0.646	Available	10000
French momentum deciles, rank-threshold panel	0.300	0.419	0.434	0.024	0.733	Available	10000
French momentum deciles, rank-threshold panel	0.500	0.419	0.434	0.024	0.733	Available	10000
French momentum deciles, rank-threshold panel	0.700	0.447	0.447	1.11e-16	N/A	Permutation null degenerate	10000
French momentum deciles, rank-threshold panel	0.900	0.497	0.404	0.042	0.015	Available	10000
French WML decile benchmark	0.900	0.497	-0.001	0.098	1.00e-04	Available	10000
French 25 Size/BM, skipped-momentum panel	0.000	0.040	0.080	0.017	0.991	Available	10000
French 25 Size/BM, skipped-momentum panel	0.300	0.034	0.097	0.023	0.997	Available	10000
French 25 Size/BM, skipped-momentum panel	0.500	0.058	0.104	0.029	0.943	Available	10000
French 25 Size/BM, skipped-momentum panel	0.700	0.087	0.137	0.041	0.892	Available	10000
French 25 Size/BM, skipped-momentum panel	0.900	0.054	0.119	0.069	0.832	Available	10000

Note: A near-zero permutation-null standard deviation is reported as N/A because the placebo is uninformative for that threshold, not calibrated evidence for or against predictability.

Table 14: HAC bandwidth, prewhitening, and fixed-b robustness diagnostics.

Candidate	Diagnostic	SR	SE	p+
AQR Betting Against Beta, equity factors	HAC K=auto	0.741	0.103	3.08e-13
AQR Betting Against Beta, equity factors	HAC K=21	0.741	0.119	2.04e-10
AQR Betting Against Beta, equity factors	HAC K=63	0.741	0.130	6.06e-09
AQR Betting Against Beta, equity factors	prewhite	0.741	0.104	4.38e-13
AQR Betting Against Beta, equity factors	fixed-b 0.1	0.741	0.166	0.002
AQR HML Devil	HAC K=auto	0.336	0.126	0.004
AQR HML Devil	HAC K=21	0.336	0.128	0.004
AQR HML Devil	HAC K=63	0.336	0.137	0.007
AQR HML Devil	prewhite	0.336	0.111	0.001
AQR HML Devil	fixed-b 0.1	0.336	0.110	0.004
AQR Quality Minus Junk	HAC K=auto	0.536	0.131	2.26e-05
AQR Quality Minus Junk	HAC K=21	0.536	0.140	6.33e-05
AQR Quality Minus Junk	HAC K=63	0.536	0.147	1.32e-04
AQR Quality Minus Junk	prewhite	0.536	0.128	1.40e-05
AQR Quality Minus Junk	fixed-b 0.1	0.536	0.151	0.002
AQR Value and Momentum Everywhere	HAC K=auto	0.573	0.159	1.62e-04
AQR Value and Momentum Everywhere	HAC K=21	0.573	0.155	1.14e-04
AQR Value and Momentum Everywhere	HAC K=63	0.573	0.144	3.41e-05
AQR Value and Momentum Everywhere	prewhite	0.573	0.158	1.42e-04
AQR Value and Momentum Everywhere	fixed-b 0.1	0.573	0.143	0.004
French momentum deciles, rank-threshold panel	HAC K=auto	0.447	0.118	7.54e-05
French momentum deciles, rank-threshold panel	HAC K=21	0.447	0.120	9.94e-05
French momentum deciles, rank-threshold panel	HAC K=63	0.447	0.122	1.21e-04
French momentum deciles, rank-threshold panel	prewhite	0.447	0.114	4.18e-05
French momentum deciles, rank-threshold panel	fixed-b 0.1	0.447	0.143	0.004
French WML decile benchmark	HAC K=auto	0.497	0.113	5.53e-06
French WML decile benchmark	HAC K=21	0.497	0.116	9.62e-06
French WML decile benchmark	HAC K=63	0.497	0.117	1.01e-05
French WML decile benchmark	prewhite	0.497	0.111	3.99e-06
French WML decile benchmark	fixed-b 0.1	0.497	0.147	0.002
French 25 Size/BM, skipped-momentum panel	HAC K=auto	0.058	0.260	0.412
French 25 Size/BM, skipped-momentum panel	HAC K=21	0.058	0.249	0.409
French 25 Size/BM, skipped-momentum panel	HAC K=63	0.058	0.231	0.401
French 25 Size/BM, skipped-momentum panel	prewhite	0.058	0.255	0.410
French 25 Size/BM, skipped-momentum panel	fixed-b 0.1	0.058	0.172	0.360

Table 15: Composite gate sensitivity by alpha threshold.

alpha	candidates	full passes	applicable passes
0.010	8	1	6
0.050	8	1	6

Table 16: Holdout and subperiod Sharpe diagnostics.

Candidate	window	SR	n
AQR Betting Against Beta, equity factors	train_70pct	0.887	17696
AQR Betting Against Beta, equity factors	holdout_30pct	0.531	7585
AQR Betting Against Beta, equity factors	quarter_min	0.280	6320
AQR Betting Against Beta, equity factors	quarter_max	1.270	6320
AQR HML Devil	train_70pct	0.465	18613
AQR HML Devil	holdout_30pct	0.120	7977
AQR HML Devil	quarter_min	0.273	6647
AQR HML Devil	quarter_max	0.445	6647
AQR Quality Minus Junk	train_70pct	0.794	12376
AQR Quality Minus Junk	holdout_30pct	0.305	5305
AQR Quality Minus Junk	quarter_min	0.174	4421
AQR Quality Minus Junk	quarter_max	1.083	4420
AQR Value and Momentum Everywhere	train_70pct	0.626	455
AQR Value and Momentum Everywhere	holdout_30pct	0.431	196
AQR Value and Momentum Everywhere	quarter_min	0.372	163
AQR Value and Momentum Everywhere	quarter_max	0.941	163
French momentum deciles, rank-threshold panel	train_70pct	0.615	18292
French momentum deciles, rank-threshold panel	holdout_30pct	0.229	7839
French momentum deciles, rank-threshold panel	quarter_min	0.171	6532
French momentum deciles, rank-threshold panel	quarter_max	1.280	6533
French WML decile benchmark	train_70pct	0.641	18292
French WML decile benchmark	holdout_30pct	0.310	7839
French WML decile benchmark	quarter_min	0.224	6533
French WML decile benchmark	quarter_max	1.370	6533
French 25 Size/BM, skipped-momentum panel	train_70pct	0.088	2428
French 25 Size/BM, skipped-momentum panel	holdout_30pct	-0.007	1049
French 25 Size/BM, skipped-momentum panel	quarter_min	-0.581	853
French 25 Size/BM, skipped-momentum panel	quarter_max	0.210	875

Table 17: Turnover-scaled cost sensitivity and break-even cost.

Candidate	cost bps	turnover	gross SR	net SR	break-even bps
AQR Betting Against Beta, equity factors	0	0.000	0.741	0.741	
AQR Betting Against Beta, equity factors	1	0.000	0.741	0.741	
AQR Betting Against Beta, equity factors	5	0.000	0.741	0.741	
AQR Betting Against Beta, equity factors	10	0.000	0.741	0.741	
AQR Betting Against Beta, equity factors	25	0.000	0.741	0.741	
AQR HML Devil	0	0.000	0.336	0.336	
AQR HML Devil	1	0.000	0.336	0.336	
AQR HML Devil	5	0.000	0.336	0.336	
AQR HML Devil	10	0.000	0.336	0.336	
AQR HML Devil	25	0.000	0.336	0.336	
AQR Quality Minus Junk	0	0.000	0.536	0.536	
AQR Quality Minus Junk	1	0.000	0.536	0.536	
AQR Quality Minus Junk	5	0.000	0.536	0.536	
AQR Quality Minus Junk	10	0.000	0.536	0.536	
AQR Quality Minus Junk	25	0.000	0.536	0.536	
AQR Value and Momentum Everywhere	0	0.000	0.572	0.572	
AQR Value and Momentum Everywhere	1	0.000	0.572	0.572	
AQR Value and Momentum Everywhere	5	0.000	0.572	0.572	
AQR Value and Momentum Everywhere	10	0.000	0.572	0.572	
AQR Value and Momentum Everywhere	25	0.000	0.572	0.572	
French momentum deciles, rank-threshold panel	0	1.91e-05	0.447	0.447	85893.000
French momentum deciles, rank-threshold panel	1	1.91e-05	0.447	0.447	85893.000
French momentum deciles, rank-threshold panel	5	1.91e-05	0.447	0.447	85893.000
French momentum deciles, rank-threshold panel	10	1.91e-05	0.447	0.447	85893.000
French momentum deciles, rank-threshold panel	25	1.91e-05	0.447	0.447	85893.000
French WML decile benchmark	0	1.91e-05	0.497	0.497	122232.000
French WML decile benchmark	1	1.91e-05	0.497	0.497	122232.000
French WML decile benchmark	5	1.91e-05	0.497	0.497	122232.000
French WML decile benchmark	10	1.91e-05	0.497	0.497	122232.000
French WML decile benchmark	25	1.91e-05	0.497	0.497	122232.000
French 25 Size/BM, skipped-momentum panel	0	0.169	0.058	0.058	2.683
French 25 Size/BM, skipped-momentum panel	1	0.169	0.058	0.036	2.683
French 25 Size/BM, skipped-momentum panel	5	0.169	0.058	-0.050	2.683
French 25 Size/BM, skipped-momentum panel	10	0.169	0.058	-0.157	2.683
French 25 Size/BM, skipped-momentum panel	25	0.169	0.058	-0.480	2.683

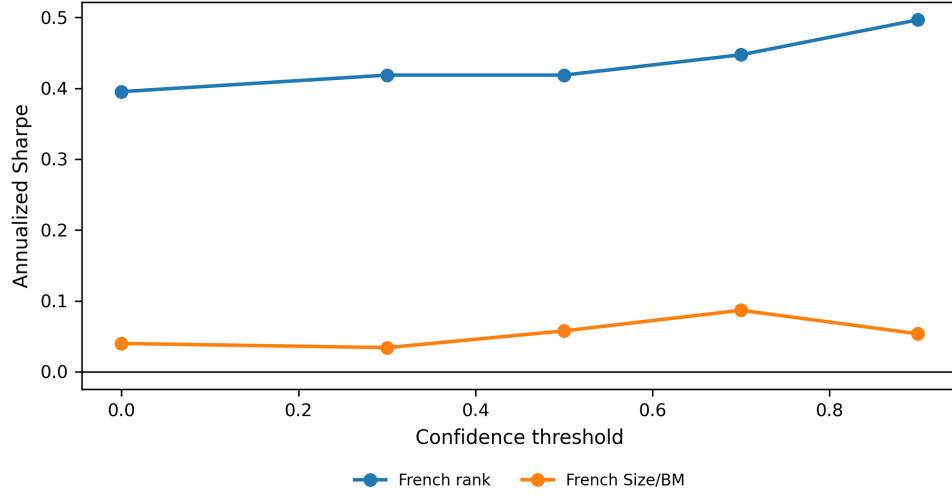


Figure 1: Threshold-profile diagnostics for public candidate panels; each line shows how annualized Sharpe changes when the registry-defined confidence threshold is varied.

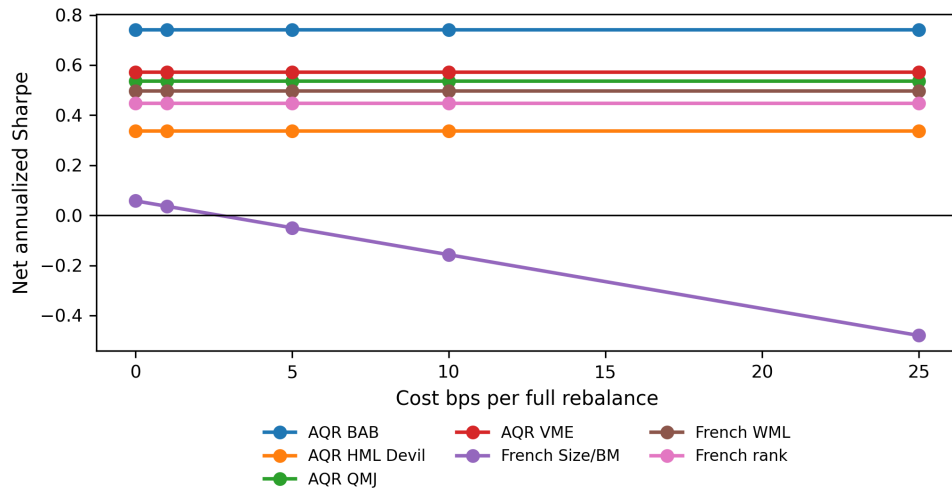


Figure 2: Turnover-scaled cost sensitivity at primary candidate thresholds; positive values indicate that the candidate remains above zero after the displayed linear cost stress.

8 Full Monte Carlo Artifacts

The simulation tables target the date-level Sharpe claim. Row-naive coverage is therefore reported against that date-level target, not against a separate row-distribution target. The design-sweep table is a stress diagnostic over same-date correlation and selected-count pressure under a zero-edge null. The main concern in this table group is size distortion: a method that looks powerful because it overrejects under the null is not acceptable for confirmatory evaluation.

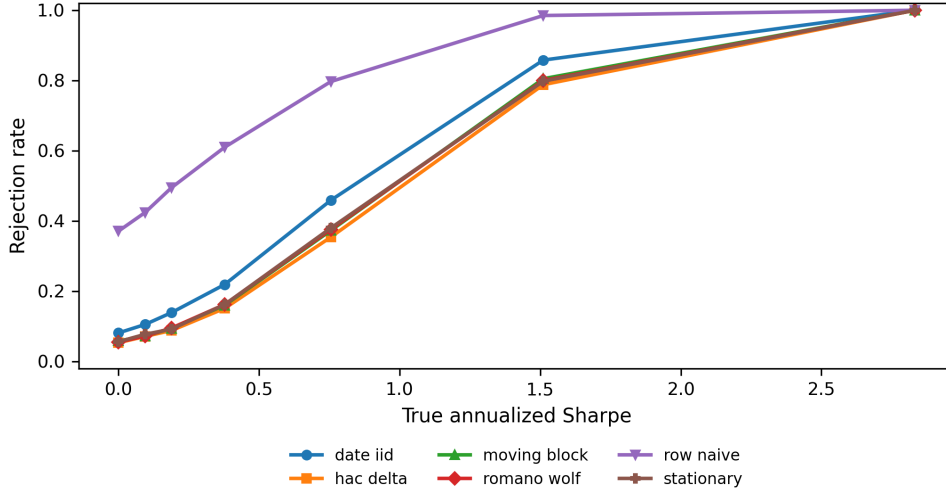


Figure 3: Monte Carlo rejection rates by true annualized Sharpe; the figure separates genuine power from methods that already overreject under zero-edge null designs.

Table 18: Monte Carlo and resampling settings.

Setting	Value
Monte Carlo replications	1000
Bootstrap draws	5000
Permutation draws	10000
Main null panel sizes	T=1000, N=50
Cross-sectional correlation grid	0.000, 0.200, 0.250, 0.300, 0.350, 0.400, 0.500
AR(1) grid	0.000, 0.150, 0.200, 0.300, 0.500, 0.700
GARCH parameter pairs	(0.050, 0.900); (0.100, 0.850); (0.150, 0.800)
HAC bandwidth rule	automatic Bartlett, with fixed-lag sensitivity checks
Moving-block length	ceil(T to the 1/5 times max(D,1)), capped at floor(T/3)
Stationary mean block	same length as moving-block rule
Dependence factor D	(2 rho1/(1-rho1)) to the 2/3 for positive AR(1) rho1; otherwise 1
Seed policy	seed-indexed replications from campaign runner

Table 19: Simulation DGP parameter grid.

DGP	Name	T	N	mu	sigma	AR1	rho	GARCH a	GARCH b	K	regime
01_iid	IID baseline	1000	50	4.00e-04	0.015	0.000	0.000	0.000	0.000	1	no
02_ser_mild	Serial mild	1000	50	4.00e-04	0.015	0.200	0.000	0.000	0.000	1	no
03_ser_strong	Serial strong	1000	50	4.00e-04	0.015	0.500	0.000	0.000	0.000	1	no
04_xs_mild	Cross mild	1000	50	4.00e-04	0.015	0.000	0.200	0.000	0.000	1	no
05_xs_strong	Cross strong	1000	50	4.00e-04	0.015	0.000	0.500	0.000	0.000	1	no
06_both_mod	Both moderate	1000	50	4.00e-04	0.015	0.200	0.200	0.000	0.000	1	no
07_both_strong	Both strong	1000	50	4.00e-04	0.015	0.500	0.500	0.000	0.000	1	no
08_realistic	Realistic	2500	100	4.00e-04	0.015	0.150	0.350	0.000	0.000	1	no
09_garch_mild	GARCH mild	1000	50	4.00e-04	0.015	0.200	0.200	0.050	0.900	1	no
10_garch_strong	GARCH strong	1000	50	4.00e-04	0.015	0.300	0.400	0.100	0.850	1	no
11_multifactor	Multi-factor K=3	1000	50	4.00e-04	0.015	0.200	0.300	0.000	0.000	3	no
12_regime	Regime-switching	2000	50	4.00e-04	0.015	0.150	0.250	0.000	0.000	1	yes
13_high_ar1	High-persistence AR(1)	1000	50	4.00e-04	0.015	0.700	0.300	0.000	0.000	1	no
14_garch_highvol	High-volatility GARCH	1000	50	4.00e-04	0.015	0.200	0.300	0.150	0.800	1	no

Table 20: Coverage of the date-level Sharpe target by alternative interval procedures.

DGP	blocked	hac_delta	iid	row_naive	stationary
01_iid	0.947	0.952	0.951	0.000	0.947
02_ser_mild	0.938	0.943	0.898	0.000	0.936
03_ser_strong	0.921	0.934	0.734	0.000	0.919
04_xs_mild	0.950	0.950	0.952	0.073	0.950
05_xs_strong	0.934	0.938	0.942	0.262	0.938
06_both_mod	0.937	0.939	0.898	0.088	0.936
07_both_strong	0.934	0.943	0.771	0.178	0.935
08_realistic	0.937	0.941	0.901	0.107	0.937
09_garch_mild	0.926	0.940	0.877	0.110	0.930
10_garch_strong	0.926	0.948	0.862	0.199	0.927
11_multifactor	0.937	0.947	0.895	0.225	0.940
12_regime	0.929	0.934	0.890	0.306	0.929
13_high_ar1	0.899	0.933	0.562	0.137	0.907
14_garch_highvol	0.940	0.948	0.895	0.193	0.941

Note: Row-naive intervals are evaluated against the economically relevant date-level Sharpe target, not against their own row-distribution Sharpe target. Low row-naive coverage therefore records target mismatch and false precision at the portfolio boundary.

Table 21: Row-naive interval behavior against the date-level Sharpe target.

DGP	Name	row coverage	bias vs date SR	CI width	mean SE	n
01_iid	IID baseline	0.000	-2.567	0.278	0.071	1000
02_ser_mild	Serial mild	0.000	-2.567	0.278	0.071	1000
03_ser_strong	Serial strong	0.000	-2.568	0.278	0.071	1000
04_xs_mild	Cross mild	0.073	-0.474	0.278	0.071	1000
05_xs_strong	Cross strong	0.262	-0.153	0.278	0.071	1000
06_both_mod	Both moderate	0.088	-0.503	0.278	0.071	1000
07_both_strong	Both strong	0.178	-0.145	0.278	0.071	1000
08_realistic	Realistic	0.107	-0.277	0.124	0.032	1000
09_garch_mild	GARCH mild	0.110	-0.473	0.278	0.071	1000
10_garch_strong	GARCH strong	0.199	-0.268	0.278	0.071	1000
11_multifactor	Multi-factor K=3	0.225	-0.327	0.278	0.071	1000
12_regime	Regime-switching	0.306	-0.092	0.197	0.050	1000
13_high_ar1	High-persistence AR(1)	0.137	-0.338	0.278	0.071	1000
14_garch_highvol	High-volatility GARCH	0.193	-0.325	0.278	0.071	1000

Table 22: Target-boundary design sweeps over same-date correlation and selected-count pressure.

design	method	rej.	SE	bias	mean interval SE	N	rho	n
rho=0.5, N=50	row_naive	0.409	0.016	0.004	0.071	50	0.500	1000
rho=0.5, N=50	hac_delta	0.047	0.007	0.005	0.597	50	0.500	1000
rho=0.1, N=50	row_naive	0.310	0.015	0.011	0.071	50	0.100	1000
rho=0.1, N=50	hac_delta	0.064	0.008	0.031	0.598	50	0.100	1000
rho=0.0, N=50	row_naive	0.097	0.009	0.002	0.071	50	0.000	1000
rho=0.0, N=50	hac_delta	0.056	0.007	0.017	0.598	50	0.000	1000
rho=0.3, N=50	row_naive	0.354	0.015	3.82e-04	0.071	50	0.300	1000
rho=0.3, N=50	hac_delta	0.044	0.006	9.09e-05	0.599	50	0.300	1000
N=5, rho=0.35	row_naive	0.192	0.012	-0.013	0.225	5	0.350	1000
N=5, rho=0.35	hac_delta	0.058	0.007	-0.018	0.598	5	0.350	1000
N=25, rho=0.35	row_naive	0.308	0.015	-0.016	0.100	25	0.350	1000
N=25, rho=0.35	hac_delta	0.052	0.007	-0.026	0.600	25	0.350	1000
rho=0.7, N=50	row_naive	0.408	0.016	0.004	0.071	50	0.700	1000
rho=0.7, N=50	hac_delta	0.060	0.008	0.004	0.599	50	0.700	1000
N=50, rho=0.35	row_naive	0.368	0.015	-1.05e-04	0.071	50	0.350	1000
N=50, rho=0.35	hac_delta	0.071	0.008	-5.58e-05	0.597	50	0.350	1000
N=100, rho=0.35	row_naive	0.421	0.016	-0.002	0.050	100	0.350	1000
N=100, rho=0.35	hac_delta	0.056	0.007	-0.003	0.598	100	0.350	1000

Table 23: Null rejection rates at nominal 5 percent size.

dgp	date_iid	hac_delta	moving_block	romano_wolf	row_naive	stationary
null_dep	0.094	0.066	0.070	0.068	0.352	0.067
null_garch	0.098	0.063	0.070	0.067	0.351	0.068
null_iid	0.060	0.060	0.061	0.060	0.062	0.062

Table 24: Rejection rates across true annualized Sharpe values.

index	true_sharpe	date_iid	hac_delta	moving_block	romano_wolf	row_naive	stationary
0.000	0.000	0.081	0.053	0.058	0.055	0.371	0.056
1.000	0.094	0.105	0.071	0.072	0.072	0.424	0.077
2.000	0.189	0.139	0.088	0.094	0.095	0.495	0.092
3.000	0.378	0.219	0.151	0.160	0.162	0.610	0.162
4.000	0.755	0.459	0.354	0.373	0.376	0.797	0.380
5.000	1.511	0.858	0.788	0.805	0.800	0.985	0.798
6.000	2.833	1.000	1.000	1.000	1.000	1.000	1.000

9 Robustness and Registry Notes

The public registry is frozen in `code/campaign_registry.json`. It records loaders, public source metadata, threshold menus, primary thresholds, and lookback/skip parameters. Changing a source, lookback, threshold, cost assumption, or eligibility rule after seeing results creates a new researcher menu and must be reported as exploratory. Rendered artifacts in the submitted package use the following SHA256 identifiers:

```
registry code/campaign_registry.json:
3a616f119f3fe2603b0663e3f68c499496c37a1781f3003e82814c9269a16259
generated campaign metadata campaign_metadata.json:
5cb21ac2cb87cb74a26090a66b28dac6890f76dd1d4bd4391ea6290267810dff
```

Stationarity diagnostics, holdout splits, HAC bandwidth sweeps, prewhitened HAC, fixed-fraction HAC, moving-block bootstrap, and stationary bootstrap results are secondary robustness checks. They are used to identify fragile claims, not to replace the main date-return decision boundary.